

Global Superstore

Sales & Profit Analysis — Power BI Dashboard

1. Project Overview

This project looks at sales data from a global superstore. The store sells products in many countries and markets, like US, EU, APAC, LATAM, EMEA, Africa, and Canada.

The goal is to understand sales, profit, customers, products, and shipping performance. The data was cleaned in Power Query and connected in a Power BI data model. Then a dashboard was built to show the results.

2. Data Source

The data comes from Kaggle:

- Dataset name: Global Super Store Dataset
- Link: <https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset/data>

The dataset has order details, customers, products, locations, sales, profit, discount, and shipping information.

3. Business Questions

These are the original business questions from the dataset:

Customer Analysis

- Profile the customers based on their frequency of purchase - calculate frequency of purchase for each customer
- Do the high frequent customers contribute more revenue?
- Are they also profitable - what is the profit margin across the buckets?
- Which customer segment is most profitable in each year?
- How are the customers distributed across the countries?

Product Analysis

- Which country has the top sales?
- Which are the top 5 profit-making product types on a yearly basis?
- What is the average delivery time across the countries?

Notes on How Some Questions Were Answered

- Country-level sales and delivery time: shown at Market level by default. Country-level detail is available using Drill Down (Market → Country) on the market charts.
- Top 5 profit-making products by year: use the Year slicer on the Product Analysis page together with the Profit by Product chart.

4. Data Model

The data is organized using a star schema. One central fact table holds the sales numbers. Four dimension tables give more details.

Naming issues (spelling mistakes and inconsistent column names in the dashboard and data model) were reviewed and cleaned up.

Table	What it holds
fact_sales	Sales, profit, quantity, discount — one row per order line
dim_customer	Customer name, segment, frequency (Loyal, Frequent, Repeat, Low Frequency)
dim_product	Category, sub-category, product name
dim_location	Country, region, market
dim_order_details	Ship mode, order priority
dim_date	Day, month, year — for time analysis

5. Dashboard Pages

The dashboard has 5 pages. Each page has filters on the left (Date, Segment, Market, Ship Mode).

5.1 Overview

A summary page with the main numbers:

- Total Sales: 12.64M
- Total Profit: 1.47M
- Total Orders: 25K
- Total Active Customers: 2K

It also shows sales and profit by year, by sub-category, and by market.

5.2 Customer Analysis

This page shows customer behavior:

- Active customers and total customers
- Profit margin: 11.61% | Retention rate: 53.31%
- Customer types: Loyal, Frequent, Repeat, Low Frequency
- Top customers by sales and profit
- Profit share by segment (Consumer, Corporate, Home Office)

5.3 Product Analysis

This page shows which products and categories perform best:

- Top products by profit (example: Canon imageCLASS)
- Profit share by category: Technology, Office Supplies, Furniture

- Sales, profit, and average discount by sub-category
- Note: Tables sub-category has a negative profit (-64,083)

5.4 Location Analysis

This page shows shipping and market performance:

- Average days to ship: 3.97 days
- Average shipping cost: 26.38
- Shipping cost by ship mode (Same Day, First, Second, Standard Class)
- Total sales and profit by market

5.5 MoM Performance

This page compares two months side by side, broken down by sub-category:

- Sales MoM %: -9.39%
- Profit MoM %: -25.36%
- Orders MoM %: +1.49%
- Helps spot which sub-categories are going up or down each month

6. Key Insights

- APAC and EU are the top markets for sales and profit.
- Technology is the most profitable category (45.23% of profit).
- Tables is the only sub-category losing money.
- Frequent Customers bring the highest total profit (879K).
- Profit dropped faster than sales this month, so margins are getting weaker.
- Same Day shipping is the fastest but has a lower cost share than Standard Class.

7. Tools Used

- Power Query — to clean and shape the data
- Power BI — to build the data model and dashboard
- DAX — to calculate measures like MoM % and Retention Rate

8. Recommendations

- Investigate why the Tables sub-category is losing money - check if the discount level is too high for this sub-category.
- Look into why profit dropped faster than sales this month (-25.36% vs -9.39%) - this may point to rising costs or heavier discounting.
- Focus retention efforts on Low Frequency customers, since they make up the largest customer group but have the lowest profit margin (9.31%).